LESSON 5.4

Apply Area Concepts to Solve Problems

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a composite figure by decomposing it into basic shapes, including a regular polygon split into triangles, and adding or subtracting the areas.

WHY THIS MATTERS Your architecture firm is calculating the floor area of an L-shaped room for a client's new home. The contractor needs the total area to order the correct amount of flooring. The room can be split into two rectangles: one is 12 feet by 8 feet and the other is 6 feet by 5 feet.

REMEMBER

Apply Area Concepts to Solve Problems

Area of Composite Figures A composite figure is a shape made of two or more simple shapes joined together. We decompose it into basic shapes, find each area, then add or subtract.

1. Decompose: Split the irregular shape into basic figures (rectangles, triangles, trapezoids).
2. Calculate: Find the area of each individual sub-figure.
3. Combine: Add the areas together (or subtract if a piece is removed/cut out)

Words: Composite Figure *(Figura compuesta)* · Decompose *(Descomponer)* · Add *(Sumar)* · Subtract *(Restar)* · Formula *(Fórmula)*

1 **Circle the letter of the best answer. Show how you know.**

In the formula $A = \frac{1}{2}(b_1 + b_2) \times h$, what do b_1 and b_2 stand for?

(A) The two slanted sides
(B) The base and the height
(C) The perimeter and the height
(D) The two parallel sides

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

What is the area of a trapezoid with bases 6 cm and 10 cm and height 4 cm?

(A) 20 square cm
(B) 32 square cm
(C) 64 square cm
(D) 240 square cm

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

What is the area of a trapezoid with bases 5 cm and 9 cm and height 6 cm?

(A) 20 square cm
(B) 42 square cm
(C) 84 square cm
(D) 270 square cm

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

A trapezoid has an area of 30 square feet with bases of 4 ft and 8 ft. What is its height?

(A) 2.5 feet
(B) 5 feet
(C) 6 feet
(D) 10 feet

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

An architect is ordering carpet for a T-shaped conference room. The top of the T is 14 feet by 4 feet and the stem is 6 feet by 10 feet. Carpet costs $5 per square foot. What is the area of the top of the T?

(A) 18 sq ft
(B) 36 sq ft
(C) 56 sq ft
(D) 60 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

6 **Circle the letter of the best answer. Show how you know.**

An architect is ordering carpet for a T-shaped conference room. The top of the T is 14 feet by 4 feet and the stem is 6 feet by 10 feet. Carpet costs $5 per square foot. What is the total area of the T-shaped room?

(A) 96 sq ft
(B) 104 sq ft
(C) 116 sq ft
(D) 840 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

For the U-shaped pool, why did we subtract instead of add? When do you add areas and when do you subtract?

SENTENCE STARTER I ___ the areas because the composite figure is made by ___ a piece from a larger shape, so the total area is ___ - ___ = ___ square feet.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.4

Apply Area Concepts to Solve Problems

Set B · Version B · Second form · core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a composite figure by decomposing it into basic shapes, including a regular polygon split into triangles, and adding or subtracting the areas.

WHY THIS MATTERS Your architecture firm is calculating the floor area of an L-shaped room for a client's new home. The contractor needs the total area to order the correct amount of flooring. The room can be split into two rectangles: one is 12 feet by 8 feet and the other is 6 feet by 5 feet.

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Plan 3 — Survey the site. Drag each part of a floor plan into the correct group: do you ADD it to the total or SUBTRACT it from a larger shape?

Groups: ADD to the total | SUBTRACT from a larger shape

_____ Triangular entrance canopy joined to the front wall

_____ Square skylight opening cut out of the roof

_____ Rectangular wing attached to the main hall

_____ Elevator shaft removed from the office floor

_____ Triangular gable joined on top of the rectangular wall

_____ Circular fountain basin carved out of the courtyard

2 **Write the letter of the matching item on each line.**

Plan 5 — Match each floor plan to its correct total area.

_____ House-front wall: 11x6 rectangle + triangle base 11, height 4

_____ T-shaped studio: 13x4 top + 5x9 stem

_____ Warehouse 20x9 minus loading bay 6x5

_____ L-shaped gallery: 7x7 + 4x6

(A) 88 sq ft

(B) 97 sq ft

(C) 150 sq ft

(D) 73 sq ft

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

An architect is ordering carpet for a T-shaped conference room. The top of the T is 14 feet by 4 feet and the stem is 6 feet by 10 feet. Carpet costs $5 per square foot. What will the carpet cost?

(A) $116

(B) $300

(C) $580

(D) $1,160

PER 1

4 **Circle the letter of the best answer. Show how you know.**

A 7 ft by 5 ft rectangle has a triangle (base 7 ft, height 4 ft) on top. Add the two areas. What is the total area?

(A) 14 sq ft

(B) 35 sq ft

(C) 49 sq ft

(D) 63 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

5 Circle the letter of the best answer. Show how you know.

Plan 2 — L-shaped lobby: decompose the L into two rectangles, one 9 ft by 6 ft and one 7 ft by 3 ft, then ADD. What is the total floor area?

(A) 21 sq ft
(B) 54 sq ft
(C) 63 sq ft
(D) 75 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

6 Circle the letter of the best answer. Show how you know.

Plan 4 — Design studio: the floor is a 16 ft by 12 ft rectangle, but a 5 ft by 4 ft storage closet is cut out of one corner. Find the usable floor area.

(A) 20 sq ft
(B) 172 sq ft
(C) 192 sq ft
(D) 212 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

EXPLAIN YOUR THINKING

For the U-shaped pool, why did we subtract instead of add? When do you add areas and when do you subtract?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help